

ALEKSANDROVSKOE, Russian Federation

WMO: 239550

Lat: 60.433N

Lon: 77.867E

Elev: 49

StdP: 100.74

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -40.7	(c) -37.8	(d) -43.1	(e) 0.1	(f) -40.3	(g) -40.2	(h) 0.1	(i) -37.5	(j) 7.3	(k) -15.4	(l) 6.6	(m) -14.9	(n) 1.8	(o) 290	(p) 0.666

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.9	(c) 28.0	(d) 19.4	(e) 26.3	(f) 18.6	(g) 24.5	(h) 17.8	(i) 20.6	(j) 25.7	(k) 19.6	(l) 24.4	(m) 18.6	(n) 23.2	(o) 3.0	(p) 110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	13.7	23.2	17.7	12.8	22.2	16.6	11.9	21.3	60.0	25.9	56.3	24.4	53.0	23.2	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.4	(b) 6.5	(c) 5.9	(e) -42.2	(f) 30.6	(g) 4.1	(h) 1.3	(i) -45.2	(j) 31.6	(k) -47.5	(l) 32.3	(m) -49.8	(n) 33.0	(o) -52.8	(p) 34.0		
(a) 7.4	(b) 6.5	(c) 5.9	(e) -42.0	(f) 23.0	(g) 3.6	(h) 1.8	(i) -44.6	(j) 24.3	(k) -46.7	(l) 25.3	(m) -48.7	(n) 26.4	(o) -51.4	(p) 27.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.2	-21.3	-17.8	-8.2	-0.9	7.1	15.4	17.8	14.3	7.9	0.2	-11.8	-18.0		
	DBStd	15.06	9.48	8.67	7.16	6.52	5.92	5.11	3.73	3.79	4.46	5.39	9.03	9.53		
	HDD10.0	4713	972	778	565	329	129	13	1	6	93	306	656	867		
	HDD18.3	7203	1230	1011	823	578	350	114	56	135	313	562	906	1125		
	CDD10.0	631	0	0	0	1	40	176	242	139	31	2	0	0		
	CDD18.3	80	0	0	0	0	3	27	39	10	1	0	0	0		
	CDH23.3	617	0	0	0	0	35	217	289	73	4	0	0	0		
	CDH26.7	109	0	0	0	0	6	44	50	9	0	0	0	0		
	WSAvg	3.0	2.8	2.9	3.3	3.4	3.4	2.9	2.6	2.6	2.9	3.2	3.1	3.0		
	PrecAvg	501	20	17	15	26	42	79	71	78	54	43	33	24		
	PrecMax	683	36	44	37	54	82	144	158	177	111	72	70	39		
(15) Precipitation	PrecMin	322	8	3	4	8	14	21	16	20	8	14	16	7		
	PrecStd	86	7	10	8	13	19	38	38	41	24	16	13	9		
	DB	0.4	0.7	6.6	17.4	27.2	30.1	29.8	27.9	23.1	15.1	4.1	0.1			
	MCWB	-0.1	-0.3	2.2	9.7	15.9	19.4	21.6	20.1	15.3	10.2	3.1	-0.4			
	2% DB	-3.1	-1.4	3.9	12.8	23.3	27.8	28.1	25.4	19.8	11.4	1.9	-1.3			
(23) Mean Coincident Wet Bulb Temperatures	MCWB	-3.8	-2.2	0.6	7.3	13.5	18.8	20.0	18.8	14.1	8.0	1.2	-1.9			
	5% DB	-5.7	-3.6	2.3	10.1	19.8	26.0	26.5	23.2	17.2	9.3	0.9	-3.0			
	MCWB	-6.3	-4.5	0.1	5.6	12.4	18.2	19.2	17.6	12.9	7.0	0.3	-3.5			
	10% DB	-8.2	-6.2	0.8	7.8	16.5	23.8	24.8	21.0	14.8	7.3	-0.3	-5.1			
	MCWB	-8.8	-7.0	-0.8	4.4	10.9	17.2	18.4	16.6	11.7	5.7	-0.9	-5.7			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	-0.1	-0.1	2.6	10.2	16.8	21.5	23.1	21.2	16.6	10.7	3.2	-0.4			
	MCDB	0.5	0.7	5.6	15.8	25.8	26.9	28.2	26.0	20.8	14.4	3.8	-0.1			
	2% WB	-3.8	-2.1	1.2	7.9	14.6	20.0	21.1	19.6	14.7	8.8	1.3	-1.8			
	MCDB	-3.2	-1.5	3.0	11.7	21.0	25.5	26.2	23.9	18.8	10.6	1.7	-1.4			
	5% WB	-6.2	-4.5	0.3	6.2	13.1	18.9	20.0	18.3	13.4	7.3	0.4	-3.5			
(31) Mean Daily Temperature Range	MCDB	-5.7	-3.7	2.1	9.8	18.8	24.5	24.8	22.0	16.5	9.0	0.9	-3.0			
	10% WB	-8.7	-7.0	-0.8	4.4	11.3	17.8	19.0	17.0	12.0	5.8	-0.9	-5.6			
	MCDB	-8.2	-6.2	0.8	7.6	16.4	22.7	23.5	20.5	14.7	7.2	-0.4	-5.1			
	5% DB	7.5	8.1	9.3	9.3	9.3	9.1	8.9	8.5	7.4	5.4	6.4	7.2			
	MCDBR	8.1	6.9	8.6	11.3	14.1	12.4	11.6	12.2	11.2	8.0	5.0	7.3			
(36) Clear-Sky Solar Irradiance	MCWBR	8.0	6.4	6.4	7.1	7.1	5.9	5.4	6.3	6.3	5.2	4.9	7.2			
	5% WB	8.3	7.0	8.2	10.7	13.0	10.7	10.4	10.8	10.2	7.3	4.9	7.6			
	MCDBR	8.1	6.5	6.6	7.2	7.2	5.7	5.5	6.0	6.2	5.2	4.9	7.5			
	taub	0.263	0.305	0.355	0.377	0.388	0.406	0.447	0.411	0.362	0.330	0.273	0.254			
	taud	1.969	2.002	1.948	2.056	2.275	2.318	2.170	2.266	2.358	2.245	2.023	1.912			
(42) All-Sky Solar Radiation	Ebn at Noon	448	648	734	805	834	824	771	772	751	631	428	288			
	Edh at Noon	48	85	125	135	118	116	131	110	84	67	44	33			
	RadAvg	0.43	1.35	2.87	4.45	4.85	5.49	5.09	3.61	2.27	1.01	0.43	0.23			
(44) All-Sky Solar Radiation	RadStd	0.04	0.11	0.18	0.41	0.42	0.45	0.59	0.43	0.29	0.10	0.10	0.02			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page